

CSC505 – Advanced Natural Language Processing

Week-wise Course Plan Class:

MS PhD – I / III (CS/SE/AI)

Semester: Spring 2026

Teacher: Sher Muhammad Daudpota (Professor of Computer Science)

Class Timings: Every Monday 5:00 PM to 8:00 PM

Venue: Room No. 201, Academic Block-V

Week No.	Topic	Assessment Tool	Reading
1	1. Introduction to Deep Learning and Connection with NLP 2. Artificial intelligence, machine learning, and deep learning 3. Before deep learning: A brief history of machine learning 4. Why deep learning? Why now? 5. The mathematical building blocks of neural networks 6. A first look at a neural network 7. Data representations for neural network		Chapter 1,2 of Book 1
2	1. Getting started with neural networks: Classification and regression 2. Classifying movie reviews: A binary classification example 3. Classifying newswires: A multiclass classification example	Assignment # 01: Text Classification with FCNN	Chapter 4 of Book 1
3	1. Fundamentals of machine learning 2. Generalization: The goal of machine learning 3. Evaluating machine learning models 4. Improving model fit 5. Improving generalization 6. The universal workflow of machine learning 7. Define the task 8. Develop a model 9. Deploy the model		Chapter 5, 6 of Book 1
4, 5, 6	1. Deep learning for text 2. Natural language processing: The bird's eye view 3. Preparing text data 4. Two approaches for representing groups of words: Sets and sequences 5. LSTM, GRU for Text Processing 6. Pre-trained word embedding (GloVe, BERT, FastText etc) 7. The Transformer architecture 8. Beyond text classification: Sequence-to-sequence learning	Assignment # 02: Text Classification with LSTM, GRU, Pre-trained Word Embedding and Transformer	Chapter 10, 11 of Book 1
7, 8	1. Text Generation Introduction 2. GANs	Assignment # 03: Text Generation	Chapter 12 of Book 1

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	3. Introduction to LLMs		
9	Mid Term (Machine Based, Open Book)		
10	1. Introduction to agents and their world 2. Defining agents 3. Understanding the component systems of an agent 4. Examining the rise of the agent era: Why agents? 5. Peeling back the AI interface	Assignment # 4	Chapter 1 of Book 2
11	1. Harnessing the power of large language models 2. Mastering the OpenAI API 3. Exploring open source LLMs with LM Studio 4. Prompting LLMs with prompt engineering		Chapter 2 of Book 2
12	1. Engaging GPT assistants 2. Exploring GPT assistants through ChatGPT 3. Building a GPT that can do data science 4. Customizing a GPT and adding custom actions 5. Extending an assistant's knowledge using file uploads 6. Publishing your GPT	Assignment # 5	Chapter 3 of Book 2
13	1. Exploring multi-agent systems 2. Introducing multi-agent systems with AutoGen Studio 3. Exploring AutoGen 4. Group chat with agents and AutoGen 5. Building an agent crew with CrewAI	Assignment # 6	Chapter 4 of Book 2
14	1. Empowering agents with actions 2. Defining agent actions 3. Executing OpenAI functions 4. Introducing Semantic Kernel 5. Synergizing semantic and native functions 6. Semantic Kernel as an interactive service agent 7. Thinking semantically when writing semantic services	Assignment # 7	Chapter 5 of Book 2
15	1. Building autonomous assistants 2. Introducing behavior trees 3. Exploring the GPT Assistants Playground 4. Introducing agentic behavior trees 5. Building conversational autonomous multi-agents 6. Building ABTs with back chaining	Assignment # 8	Chapter 6 of Book 2
16	1. Assembling and using an agent platform 2. Introducing Nexus, not just another agent platform 3. Introducing Streamlit for chat application development 4. Developing profiles and personas for agent	Assignment # 9	Chapter 7 of Book 2

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	5. Powering the agent and understanding the agent engine 6. Giving an agent actions and tools		
17	1. Understanding agent memory and knowledge 2. Understanding retrieval in AI applications 3. The basics of retrieval augmented generation (RAG) 4. Delving into semantic search and document indexing 5. Constructing RAG with LangChain 6. Applying RAG to building agent knowledge 7. Implementing memory in agentic systems 8. Understanding memory and knowledge compression	Assignment # 10	Chapter 8 of Book 2
18	9. Mastering agent prompts with prompt flow 10. Why we need systematic prompt engineering 11. Understanding agent profiles and personas 12. Setting up your first prompt flow 13. Evaluating profiles: Rubrics and grounding 14. Understanding rubrics and grounding 15. Grounding evaluation with an LLM profile 16. Comparing profiles: Getting the perfect profile		Chapter 9 of Book 2
19	Final Exam (Machine Based, Open Book)		

Textbooks

Book 1: Deep Learning with Python, Second Edition (2021, Manning)

Book 2: AI Agents in Action -- Micheal Lanham (2025, Manning)

Marks Distribution:

Category	Category Weight
Mid Term	20%
Final Term	50%
Assignments	10%
Project Submission	20%